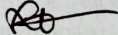


SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	DSA0301	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 1 – ANALYTICAL PROBLEM SOLVING
Weightage:	50% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:	T. Bhanu Teja	Reg. No.:	192472259
Section / Batch:	CSE - 2024	Date:	05-08-26

Code of Conduct

I T. Bhanu Teja / 192472259 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: 

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Part of Speech Tagging	Morphological decomposition of connected, connecting, and connection; identification of roots, suffixes, inflectional vs. derivational forms, and normalization for document indexing.	Q1
English Word Classes, Tagsets for English, Rule-Based Part of Speech Tagging	Morphological parsing of unhappy, happiness, and happily; identification of prefixes, suffixes, base forms, and semantic effects of derivational morphology.	Q2
Counting Words, Part of Speech Tagging, English Word Classes	Stemming and root extraction for played, player, and playing; handling grammatical variations and word formation changes for search preprocessing.	Q3
Finite-State Morphological Analysis, Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Finite-state parsing of writes, writing, and written; modeling regular and irregular verb inflections and root normalization.	Q4
English Word Classes, Rule-Based Part of Speech Tagging, Counting Words	Porter Stemmer-based processing of relational, relation, and relate; application of derivational suffix removal and stem extraction for retrieval.	Q5

Annexure A – ANALYTICAL PROBLEM SOLVING

1. Given the input words: "connected", "connecting", "connection", design a morphological analysis pipeline for a document indexing system.

Apply rules to decompose each word into root and suffix components.

Differentiate between inflectional and derivational forms within the dataset.

Normalize all variants to a common base form for efficient retrieval.

Determine the parsed structure and normalized output for each word.

2. Given the input words: "unhappy", "happiness", "happily", design a morphological parsing module for sentiment analysis.

Identify prefixes, suffixes, and base forms using rule-based decomposition.

Ensure semantic consistency across derived word forms.

Classify each transformation as inflectional or derivational.

Produce the root and morphological breakdown for each word.

3. Given the input words: "played", "player", "playing", design a stemming-based preprocessing module for a search engine.

Apply morphological rules to strip affixes and identify the base form.

Handle both grammatical variations and word formation changes.

Ensure consistent root extraction across all forms.

Determine the stem and classification for each input word.

4. Given the input words: "writes", "writing", "written", design a finite-state morphological parsing module for a text processing system.

Apply state transitions to represent suffix transformations and irregular forms in verb inflection.

Differentiate between regular and irregular inflectional patterns within the dataset.

Ensure consistent mapping of all forms to a common root representation.

Determine the state transitions, morphological breakdown, and normalized output for each word.

5. Given the input words: "relational", "relation", "relate", design a Porter Stemmer-based preprocessing module for document retrieval.

Apply stemming rules step-by-step to remove derivational suffixes and obtain base forms.

Handle variations in suffix patterns while preserving semantic consistency.

Ensure uniform root extraction across all derived forms.

Determine the stemming steps, intermediate forms, and final stem for each input word.

Rubrics for Calculation

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Q. No. / Criteria Level	Problem Understanding	Method Selection	Solution Process	Accuracy of Calculation	Interpretation of Results	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

05-08-26

C02-AT1

T. Bhanu Teja

192472259

DSA0301

Natural Language
Processing

Q1. Morphological Analysis - for Document Indexing

Input words:- Connected

Connecting

Connection

Aim, to analyze different forms of the word Connect, identify their morphological structure, classify them as inflectional or derivational, and normalize them for efficient document retrieval.

Morphological Decomposition:

<u>Word</u>	<u>Root</u>	<u>Suffix</u>	<u>Type</u>
Connected	connect	-ed	Inflectional
Connecting	connect	-ing	Inflectional
Connection	connect	-ion	Derivational

Morphological Rules:

- * -ed → Past tense
- * -ing → present Participle
- * -ion → forms a noun from a verb

Classification:

Inflectional forms

* Connected

* Connecting

→ only grammatical form changes.
Meaning remains "connect."

Derivational form

* Connection

→ Changes the word class
from verb → Noun

Normalization:

All words :-

Normalized Root

Connected
connecting
connection

→ Connect

Final Parsed structure:

Word	Parsed structure	Normalized output
Connected	Connect + ed	connect
connecting	connect + ing	connect
connection	connect + ion	connect.

Q2: morphological Parsing for Sentiment Analysis

Input words:-
*unhappy
*happiness
*happily

Aim: To analyze word formation, identify prefixes, suffixes, and base forms, and preserve semantic consistently for sentiment analysis.

Morphological Break down:-

<u>Word</u>	<u>Prefix</u>	<u>Root</u>	<u>Suffix</u>
unhappy	un	happy	-
happiness	-	happy	ness
happily	-	happy	ly

Rules: un + happy

↓
negative adjective
happi⁺ness → Noun
happy + ly → Adverb

Classification:-

<u>Word</u>	<u>Type</u>
unhappy	Derivational
happiness	Derivational
happily	Derivational

Reason:- All three change either the meaning or the grammatical category.

Semantic Consistency

Although the forms differ,
unhappy
happiness
happily

All originate from : happy

Final output:-

<u>Word</u>	<u>Breakdown</u>	<u>Root</u>
unhappy	un+happy	happy
happiness	happy+ness	happy
happily	happy+ly	happy

Conclusion:-

The parser identifies prefixes, suffixes, and the common base word while preserving semantic relationships among derived forms.

Q3:- Stemming for Search Engine:-

Input: → played
→ player
→ playing

Aim:- TO remove affixes and identify a common stem for better search performance.

Morphological Analysis:-

<u>word</u>	<u>Root</u>	<u>Affix</u>	<u>Classification</u>
Player	play	er	Derivational
played	play	ed	Inflectional
Playing	play	ing	Inflectional

Stemming,
played
↓
play
playing
↓
play
player
↓
play

Final table:-

<u>Word</u>	<u>stem</u>	<u>Classification</u>
played	play	Inflectional
player	play	Derivational
playing	play	Inflectional

Conclusion,

The stemming module extracts the common stem play, improving document retrieval by treating all forms as equivalent.

Q4. Finite - State Morphological Parsing

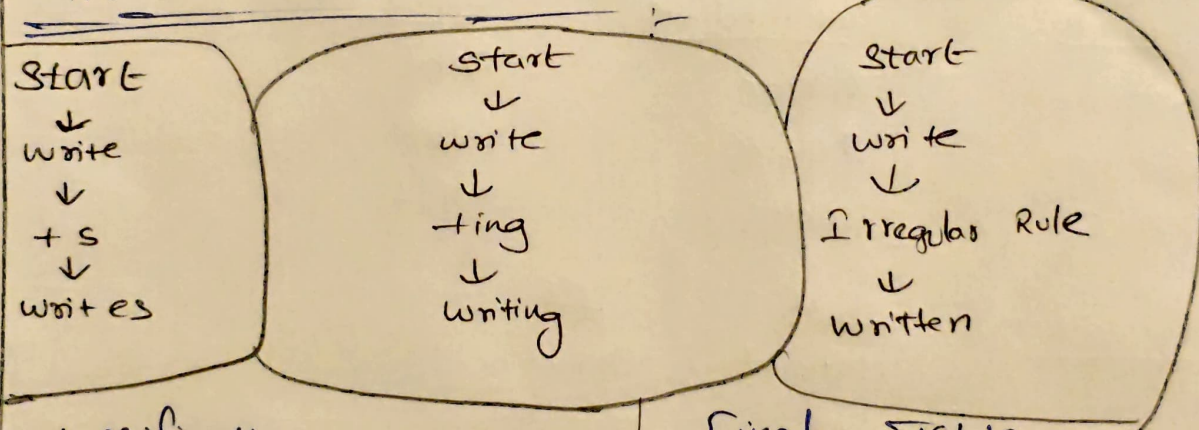
Input → * writes
→ writing
→ written

Aim To model regular and Irregular verb inflections using finite-state morphological parsing.

Morphological Breakdown

<u>word</u>	<u>Root</u>	<u>Suffix</u>	<u>Pattern</u>
writes	write	s	Regular
writing	write	ing	Regular
written	write	en	Irregular

Finite State Transitions



Classification

<u>word</u>	<u>Type</u>
writes	Regular
writing	Regular
written	Regular

Final Table

<u>word</u>	<u>State Transition</u>	<u>Root</u>
writes	write → +s	write
writing	write → +ing	write
written	write → Irregular → written	write

Conclusion

Finite State Morphological Parsing handles both regular and Irregular verb forms while mapping them to common root write.

Q5. Porter Stemmer

Input:-
→ Relational
→ Relation
→ Relate

Aim:- To apply Porter stemming for removal derivational suffixes and obtaining a common stem.

<u>Stemming</u>	Relational ↓ remove "ational" ↓ Relate	Relation ↓ remove "ion" ↓ relate
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Relate:- Already base word.

Intermediate forms:

<u>Words</u>	<u>Intermediate</u>	<u>Final stem</u>
Relational	Relate	Relat
Relation	Relate	relat
Relate	relate	relat

<u>Classification</u>	<u>word</u>	<u>Type</u>
	relational	Derivational
	relation	Derivational
	relate	Base word

<u>final output</u>	<u>word</u>	<u>Final stem</u>
	relational	relat
	relation	relat
	relate	relat

Conclusion

The Porter Stemmer removes derivational suffixes from all input words and produces the common stem relat, ensuring consistent indexing and efficient document retrieval.